Relationship between the length and the time taken for a u folded chain to unfold under the influence of gravity

1.0 Introduction

This paper is focused on assessing the whipping effect defined as a phenomena where an inextensible string under tension experiences larger lateral oscillation or vibrations due to its own elasticity and to what extent the acceleration of the end point of the chain is influenced by this effect. Fundamental physics states that a body close to the surface of the earth experiences uniform acceleration due to gravity equal to $9.8\ ms^{-1}$ although experiments have proven that a U folded chain unravels faster due to higher acceleration experienced by the end point of the chain. This situation is an example of dynamic mass systems similar to that of a rocket expelling fuel wherein the subsequent motion of the falling chain link leads to reduction of mass falling as demonstrated in Image 1.0.

This dynamic system is relevant and applicable in many real life situations such as the motion of a whip, the space tether, bungee jumping, etc and studying the situations of a metallic chain unfolding under the influence of gravity is an appropriate simplification to address and analyse more complex systems through which the following research question can be answered.

Research Question - To what extent does the length of a metallic chain (5,6,7,8,9,10,11,12 (ft)) affect the time taken for the chain to unravel (s) under the influence of gravity (9.8 ms^{-1})?

2.0 Background information

Let us first consider a metal chain suspended from a rigid support with one of its ends fixed to the rigid support and the other end freely suspended such that the chain hangs from the rigid support and is at rest. We will consider the chain and rigid support to be a system and therefore normal reaction forces felt by the support and the chain due to being in contact can be ignored as they cancel each other out in accordance with Newton's Third Law of Motion. Moreover since the chain is at rest the forces acting on the chain at any given point must be equal and the net force acting on the chain at any given point on its length must be 0. **Figure 1.0** depicts the forces acting on this hanging chain demonstrating how at a given point *C* the weight of the chain hanging below point *C* = *BC* is balanced by the tension in the chain in section *AC* of the chain. Therefore the tension at any given point on the chain depends on the weight of the segment of the sub-chain below that point.

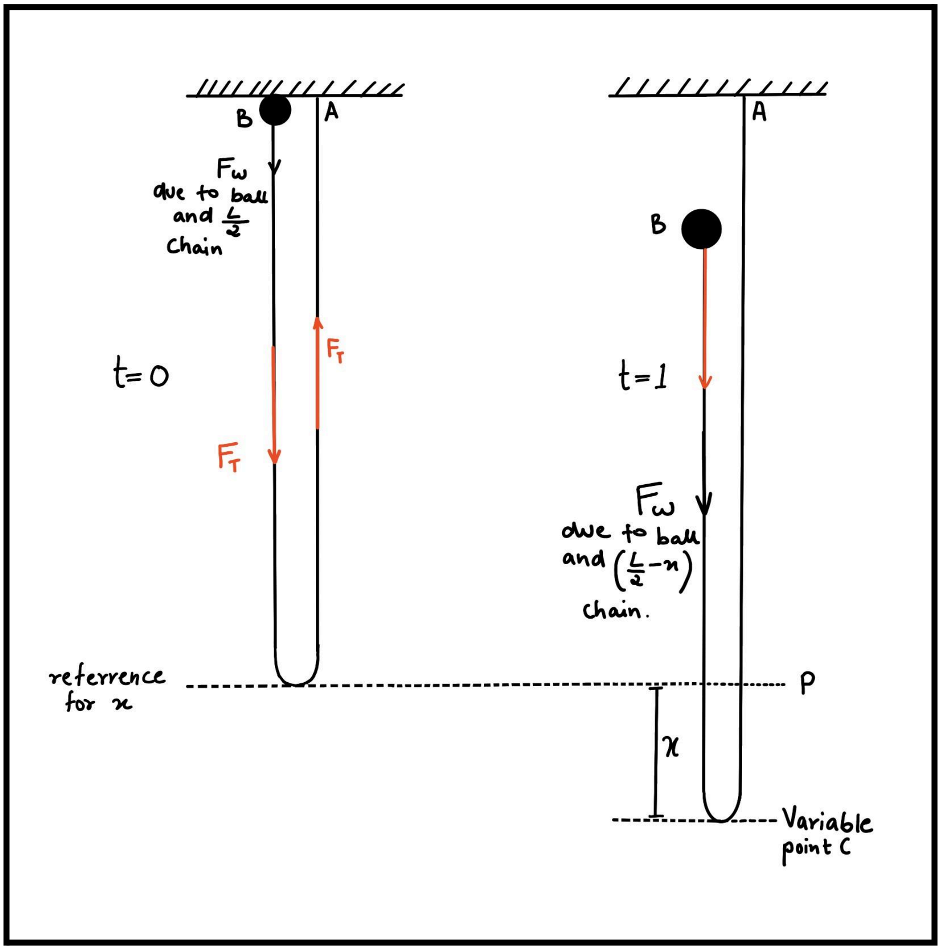


Figure 1.0 visual aid to explain the context of this experiment and is a free body diagram of the forces acting along the chain. Created by author using CGI tools.

Similarly now consider the chain *AB* where *A* is fixed to a rigid support and *B* is the loose end that is held at the same height as *A* and dropped. Let us also fix a point *P* that is the lowest point of the chain in its hanging U shaped position. Since the chain is at rest all the forces acting on the chain cancel each other out and hence the net force on the chain is zero. Let us consider that the length of the chain is *L* and hence $AP = BP = \frac{x}{2}$. A black ball of mass *M* is attached to point *B* of the chain and is held over the plane on which points A and *B* of the chain are placed. When the object is left to accelerate under gravity we will consider that small sub-chains are added to the length of the main chain *AP*. Therefore as the object falls, the length of the main chain will increase therefore we will assume that the increase in main chain AP from point *P* is *x*. Since the length of the segment of chain below fixed point *P* increases, the weight of the segment of the subchain below P will increase and therefore tension T_x in the string will increase. This relationship can be given by the following:

$$Tension\ in\ chain\ =\ weight\ of\ chain\ under\ point\ P$$

$$T\ =\ mg$$

$$Let\ us\ consider\ \lambda\ to\ be\ mass\ of\ chain\ per\ unit\ length$$

$$Therefore\ m\ =\ \lambda x$$

$$T_x\ =\ \lambda xg$$

$$Considering\ a\ ball\ of\ mass\ M\ is\ attached\ to\ end\ B\ of\ the\ chain$$

$$Net\ weight\ of\ the\ chain\ as\ it\ unravels\ =\ W_{unravelled\ sub-chain}\ +\ W_M$$

$$=\ \lambda xg\ +\ Mg$$

$$According\ to\ newton's\ second\ and\ third\ law$$

$$F\ =\ \lambda xg\ +\ Mg$$

Initially when the object is left x is minimal and hence the object experiences a force $=\ Mg$ accelerating due to gravity only but as length of unravelling sub-chain becomes more prominent the tension force acting throughout the length of the chain, mostly at the end of the chain where the object is attached changes its rate of change of velocity and experiences acceleration greater that g due to the additional factor of λxg.

In order to find a mathematical relationship between the time taken for the chain to unravel and the length of the chain we will employ the two following methods: Newton's alternate form of the second law and the principle of conservation of energy.

2.1 Newton's Law Approach

We already deduced previously that $F = \lambda xg + Mg$ **let this be Equation 1**

We can use the alternate form of Newton's Second Law of Motion given by

$$F = \frac{\Delta p}{\Delta t} = \frac{\Delta(mv)}{\Delta t}$$

Which can also be written as

$$F = \frac{dp}{dt}$$

We know that the mass of the chain and the object or $\Delta m = \lambda x + M$

Therefore $F = \frac{d}{dt}(\lambda x + M) \times v$ ***(apply product rule of differentiation)***

$$F = \frac{d}{dt}(\lambda x + M)v + (\lambda x + M)\frac{dv}{dt}$$ (***M can be removed front he first bracket as it's independent of x)***

$$F = \frac{dx}{dt}(\lambda)v + (\lambda x + M)\frac{dv}{dt}$$

$$F = \lambda v^2 + (\lambda x + M)a$$ **let this be equation 2**

Equate equation 1 and equation 2

$$\lambda xg + Mg = \lambda v^2 + (\lambda x + M)a$$

$$\frac{\lambda xg + Mg - \lambda v^2}{\lambda x + M} = a$$

$$\frac{dv}{dx}v = \frac{\lambda xg + Mg - \lambda v^2}{\lambda x + M}$$ ______________________ **(Equation 1)**

(***master differential equation that can be solved to find v(x))***

2.2 Energy Conservation approach

Under the assumption that no external non conservative forces are acting and friction and drag is negligible we can use energy conservation to deduce a relationship between length of the chain and time to unravel.

Since the U folded chain is split into two sections AP and BP the centre of mass of each section is $\frac{AP}{2}$ and $\frac{BP}{2}$ respectively as given in Figure 2.0.

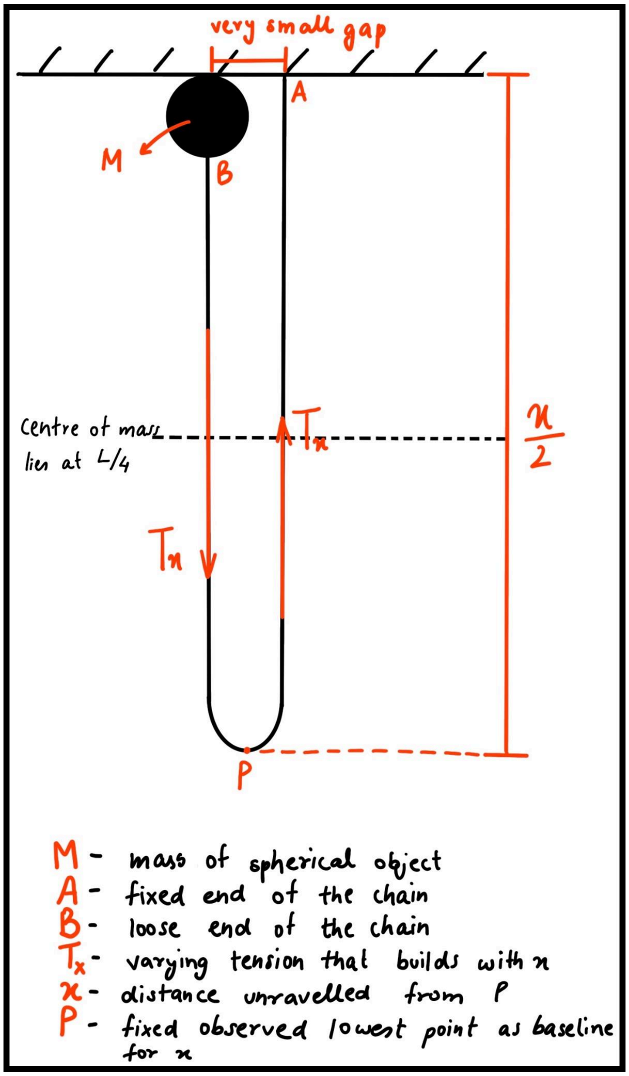


Figure 2.0 depicts the hypothetical setup to understand the energy conservation approach detailed below. Drawn by author using CGI tools.

$$\text{Since } AP = BP = \frac{x}{2}$$

therefore

$$\frac{AP}{2} = \frac{BP}{2} = \frac{x}{4} = centre\ of\ mass\ of\ the\ U\ folded\ chain$$

$$total\ potential\ energy\ of\ the\ system = PE_{object} + PE_{sub-chain}$$

$$= Mg\frac{x}{2} + M_x g\frac{x}{4}$$

As the chain unravels the chain links gain a velocity due to the conversion of potential energy to kinetic energy. The falling ball of mass M already converts its potential energy to kinetic energy but further the chain link's acceleration adds to the kinetic energy of the system.

$$the\ object\ and\ chain\ observe\ a\ displacement\ x\ time = t$$

$$For\ the\ black\ ball\ fallen\ x\ metres\ at\ time = t$$

$$-\Delta PE = \Delta KE$$

$$Mgx = \frac{1}{2}Mv^2$$

$$as\ the\ chain\ unravels\ forming\ sub\ chain\ of\ length\ x\ at\ time\ =\ t$$

$$-\ \Delta PE\ =\ \Delta KE$$

$$\lambda xg\frac{x}{2} = \frac{1}{2}\lambda xv^2$$

$therefore\ total\ kinetic\ energy\ of\ the\ system\ at$ time t

$$\left|\Delta PE_{total}\right| = \left|\Delta KE_{total}\right|$$

$$Mgx + \frac{\lambda x^2 g}{2} = \frac{1}{2}Mv^2 + \frac{1}{2}\lambda xv^2$$

$$upon\ isolating\ for\ v\ we\ get$$

$$v = \sqrt{\frac{2}{M+\lambda x}\left(Mgx + \frac{\lambda gx^2}{2}\right)}$$

$$v = \frac{dx}{dt}$$

$dt = \frac{dx}{v}$ ***(substitute v)***

$$dt = \frac{\sqrt{M+\lambda x}}{\sqrt{2Mgx+\lambda gx^2}}dx$$

$$integrate\ both\ sides$$

$t = \int_0^{\frac{x}{2}} \frac{\sqrt{M+\lambda x}}{\sqrt{2Mgx+\lambda gx^2}}dx$ ____________________**(Equation 2)**

$$Where:\ m = mass\ of\ the\ chain$$

$$M = mass\ of\ the\ sphere\ attached\ to\ the\ end\ point\ of\ the\ chain$$

$$g\ is\ acceleration\ due\ to\ gravity$$

$$t = time\ taken\ to\ unravel$$

$$x = length\ of\ the\ chain$$

These approaches however are beyond the scope of this experiment as they give more accurate answers based on the chain link density and the mass of the chain therefore in order to find a conclusive trend for the experiment we take advantage of one of the fundamental equations of motion given by

$$s = ut + \frac{1}{2}at^2$$

Since the suspension point will be at rest initial $u = 0$ thus giving us

$$s = \frac{1}{2}at^2 \qquad \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_}\textbf{(Equation 3.0)}$$

Which can be modelled to be a straight line in the form of $y = mx + c$

Where:

$$y = s$$

$$x = t^2$$

$$m = \frac{a}{2}$$

$$c = 0$$

From the above deduced relation we can plot a graph between the drop height of the sphere or the length of the chain against the time we record for the ball to reach the displacement mark or for the chain to unravel completely squared. The gradient of the graph $\times\ 2$ will hence give us the acceleration experienced by each system and plotting a graph of height/length of chain against acceleration would allow us to verify or reject the hypothesis.

3.0 Research Design

3.1- Independent Variable: Length of the metal chain (5.0, 4.5, 4.0, 3.5, 3.0, 2.5, 2.0, 1.5, 1.0 (m))

Justification: As the length of the chain increases the mass of chain links hanging under the point of suspension (the spherical ball in this case) would also increase and therefore according to the known relationship between mass and acceleration when the net force acting is constant, $F = ma$, we can determine that the suspended sphere when released will accelerate faster than the same sphere (same mass) when it'sn't attached to the chain. However the extent of this acceleration remains relatively unknown with respect to length of the chain making it suitable to assess in this experiment. **Method to manipulate:** The independent variable was varied by increasing the number of links in the chain for every increment and a measuring tape was used to measure the length of the chain for every trial and change in independent variable. **Uncertainty:** since the measuring tape used was an analog device its uncertainty was determined by dividing the least count by 2 which is $\pm\ 0.05$m.

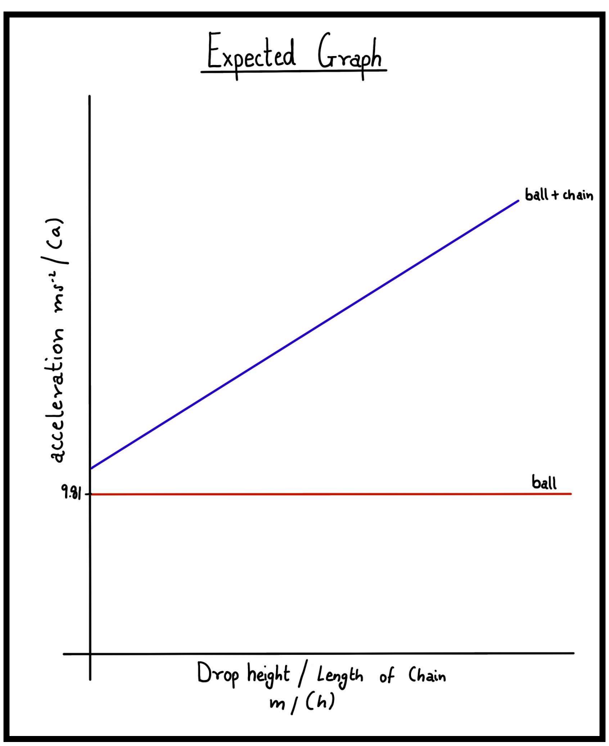
<u>3.2 Dependent Variable:</u> Time taken to unravel till point P (s)

Method to find: Since this dynamic motion of the chain unravelling is quite spontaneous the most ideal way to quantify the effect of varying the independent variable is by measuring the time taken by the chain to unravel till a fixed point. The time taken to unravel will be compared to that of a body of the same mass under free fall. A video analysis of both these motions subsequently in slow motion will give us the time taken for both objects to reach a known height which will then be plotted against the length of the chain in order to deduce a relationship based on the trend of the graph. **Method to record:** The time will be measured using a slow motion video recorded by a camera and then determined using a video analysis and tracking software. **Uncertainty:** Uncertainty was propagated due to precision of camera device and video analysis software. Research indicated its uncertainty to be $\pm$0.06 s for a 60 fps rating.

Table 2.0 is a record of the controlled variables of the experiment with justification for the same

Controlled variable	Value at which it's controlled	Why is it controlled	How to control
Drop height of the	12 meters	Changing the drop height would change the	Use the same reference point to

change		amount of potential energy stored in the sphere that would then increase the kinetic energy that will get transformed increasing the acceleration and time observed as the chain unravels	hang the chain in its initial U shaped position. Ensure that the position of the fixed end of the chain is consistent throughout all trails.
Chain link density	NA	According to the equation for potential energy where $PE = mgh$ the density of the chain would impact the mass of the chain links and therefore the chain that hangs under the point of suspension that pulls the sphere down. This would further affect the acceleration and time recorded for the unravelling of the chain.	Use the same chain and chain links throughout all trials of the experiment. Ensure that all the chain links are similar.
Mass distribution across the length of the chain	NA	The reason stated for Chain Link Density applies here as well. More importantly, irregular distribution of mass across the chain would cause the chain to unravel with oscillation due to the imbalance of mass leading to discrepancy in the reading.	Use consistent chain links throughout the length of the chain. Ensure that the mass added per increment of the independent variable is consistent.
Size of the chain links across the length of the chain	NA	Since size of the links is directly proportional to the mass of the links it must be kept constant to ensure distribution of mass and chain link density is constant as well.	By using the same chain links as that of the initial chain.



4.0 Hypothesis

Since literature review has already confirmed that an object attached to the end of a U shaped chain will experience an acceleration greater than acceleration due to gravity. Hence I hypothesize that increasing the length of the chain will take longer to unfold, however the acceleration experienced by the chain upon increasing its length will decrease the time taken to hit the ground.

Figure 3.0 hypothesized graph

Justification: As seen in **Section 2.0** the above statement can be hypothesized as the mass of the chain links under the sphere that is suspended decreases the acceleration experienced by this sphere increases as the net force (gravitational force) must remain constant.

Initially the higher mass results in higher potential energy stored in the chain and sphere that is suspended that upon unfolding gets transformed as kinetic energy that is gained in the end mostly by the sphere as the chain links unravel and come to rest transferring this energy up towards the sphere increasing its velocity.

5.0 Safety Ethics and Environment

- Cutting and adding the metal links had a certain extent of danger that could'vecaused cuts to the experimenter. Rubber gloves were worn to ensure the safety of the experimenter.
- Moreover metal shards could also be ingested or hurt the eyes of the experimenter and therefore mask and safety goggles were ensured to be worn.
- It was ensured that ethical use of resources was followed through the utilization of a set amount of resources and its conservation.
- The waste products from this experiment were disposed appropriately in dustbins ensuring the safety of the environment.

6.0 Apparatus and Setup

Sr No.	Apparatus	Quantity	Uncertainty	Percentage Uncertainty
1	long chain	5 m	± 0.0001	± 0.002
2	Extra chain links	20	NA	NA

3	Black/Blue coloured spherical mass	2	NA	NA
4	High Frame rate video recorder	1	± 0.001	Highest recorded value: $\pm$ 0.4
5	Electrical tape	1 roll	NA	NA
	Total Percentage Uncertainty			$\pm$**0.402**

In the above table the calculated total percentage uncertainty of the apparatus use is less than 5% deeming the experiment to be reliable and precise.

7.0 Methodology

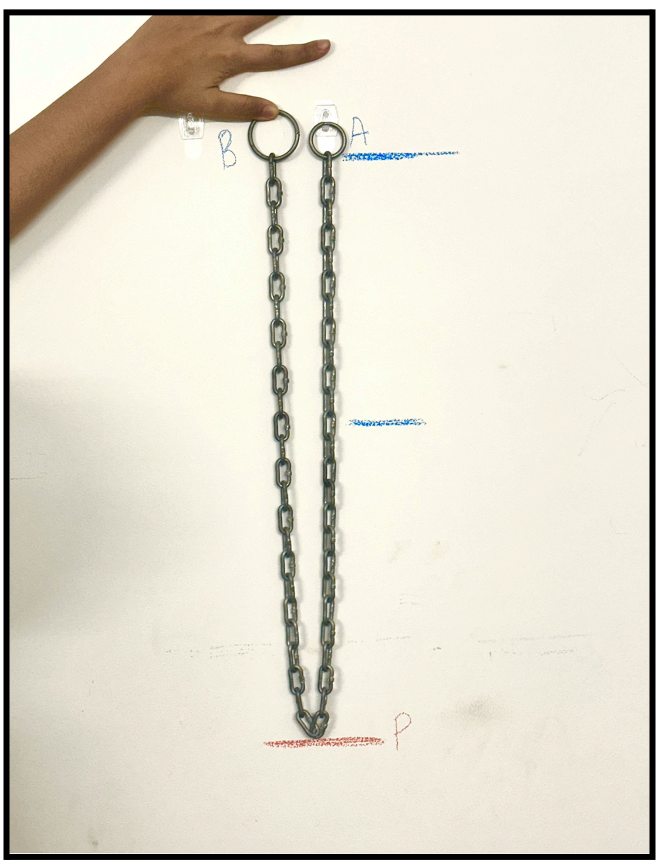


Figure 4.0 depicts the apparatus setup done for the experiment. The image to the left was taken for clarity and coherence with the background information.

1. Take a chain of length x and fix one end of the chain to the wall. Take a relatively long nail to ensure that the chain is not in contact with the wall. Mark this spot on the wall using electrical tape.
2. Mark regular intervals of 0.2 meters on the wall using the electrical tape parallel to the suspended chain at rest.
3. It must be ensured that the chain's minima, e.i. point P is a minimum distance of $\frac{x}{2}$ above the ground. In order to do this i calculated the maximum length chain i will be analysing the motion of, divided that by two and set the distance of P from the ground to be ≥ that length.
4. Let the chain freely hang from the nail and mark the point where the other end of the chain is using a brighter red tape. This will be the base point for finding the final time at which the chain has unravelled.
5. A phone camera with a tripod was placed at a distance of 2 meters from the wall with the suspended chain and was set to slow motion to analyse the motion of the unfolding chain by a factor of 8x. It must be ensured that the angle and distance from the wall of the camera in all trials are consistent and hence the wall from the black tape where the chain is nailed to the red tape (end point of the chain's motion) must be in the frame.
6. Set up a double clamp to hold the other end of the chain and the string attached to the black ball to drop it at the same time, minimizing human error, such that the initially nailed down end of the chain, the suspended end of the chain and the ball are horizontally at the same level on the wall.
7. The video recording was started and the clamps were loosened at the same time while ensuring that the chain was absolutely still before the motion of the chain unfolding.
8. Ensure that the recording is stopped only after the chain has come to rest in its unravelled state.
9. Using a video viewing software and note down the initial time and which the clamp was loosened and the chain and ball were left under the influence of gravity and the final time at which the chain and the ball reached their lowest mark (red tape on the wall).
10. Repeat this process 3 times for the same length of chain to reduce random error and eliminate outliers
11. Add 3 chain links to the original chain and repeat steps 1-11 for lengths 5.0, 4.5, 4.0, 3.5, 3.0, 2.5, 2.0, 1.5, 1.0 (m)

8.0 Data Processing

Table 1.0 is a consolidation of the data collected for the dependent variable

Length of the chain m/ (L) ± 0.05	Time taken for chain to unravel s/ (t) ± 0.06					Mean time taken for chain to unravel s/ (t) ± 0.06
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	
5.00	12.73	13.54	13.08	12.92	13.29	13.11
4.50	11.35	12.07	12.87	11.89	12.27	12.09
4.00	11.44	11.37	11.81	11.62	11.41	11.53
3.50	10.46	10.46	11.46	10.79	10.79	10.79
3.00	9.6	10.42	10.66	10.15	10.27	10.22
2.50	9.66	8.98	10.21	9.56	9.68	9.62
2.00	8.42	10.03	9.09	9.17	9.19	9.18
1.50	8.56	9.44	7.75	8.52	8.67	8.59
1.00	7.85	8.15	8.25	7.95	8.05	8.050

Table 2.0 is a conversion of the mean slow motion time to actual time by dividing by a factor of 16 and squaring the actual time.

Length of the chain m/ (L) ± 0.05	Mean time taken for chain to unravel s/ (t) ± 0.06	Actual time s/ (t) ± 0.06	Time square s^2/ (t^2) ± 0.120	Acceleration ms^{-1}/ (a)

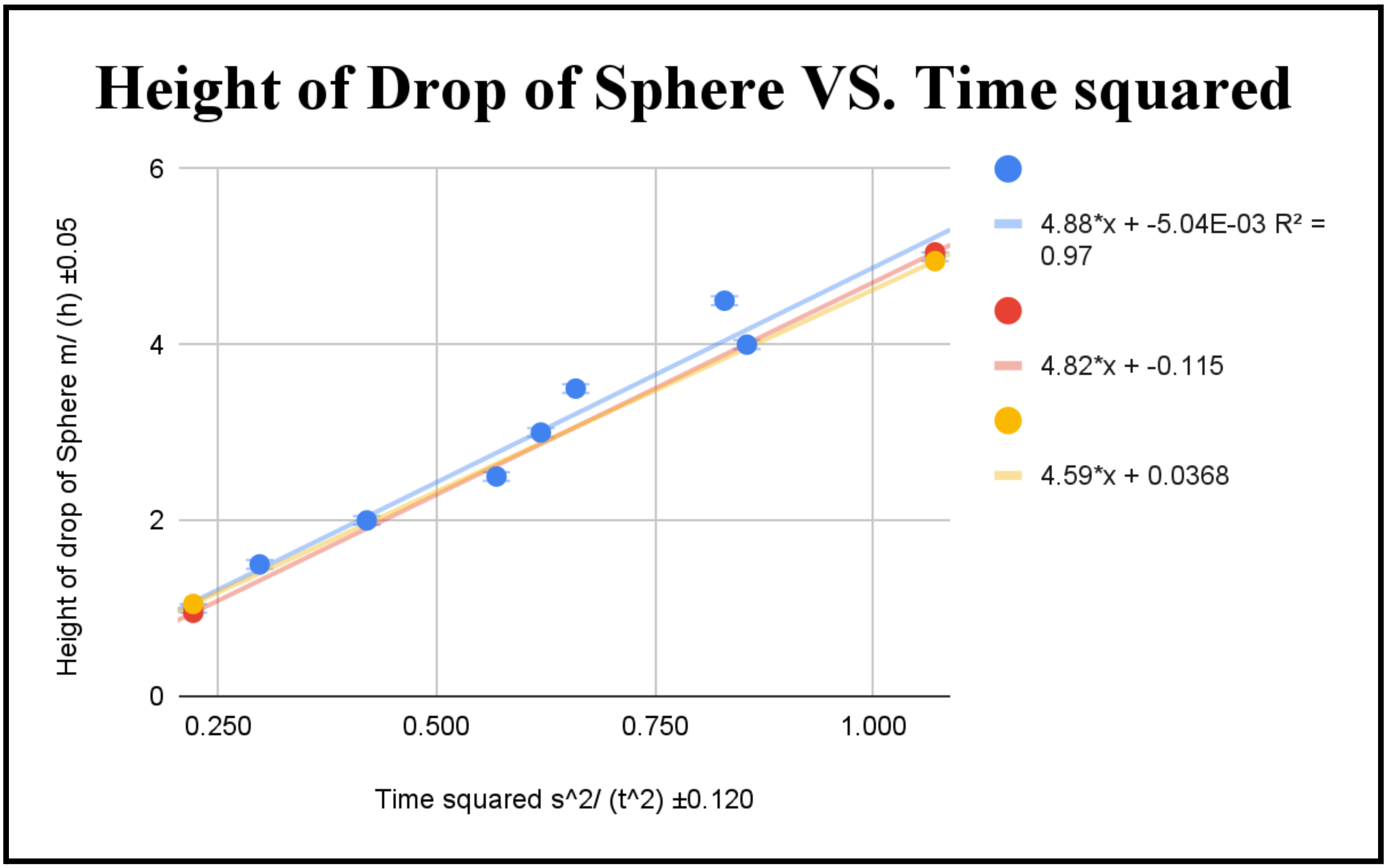
5.00	13.11	0.819	0.671	14.895
4.50	12.09	0.756	0.571	15.763
4.00	11.53	0.721	0.519	15.405
3.50	10.79	0.674	0.455	15.392
3.00	10.22	0.639	0.408	14.706
2.50	9.62	0.601	0.362	13.831
2.00	9.18	0.574	0.329	12.151
1.50	8.59	0.537	0.288	10.408
1.00	8.050	0.503	0.253	7.901

Table 3.0 shows the slow motion time and actual time taken for a uniform black sphere to reach the ground under the influence of gravity at the same independent variable increments.

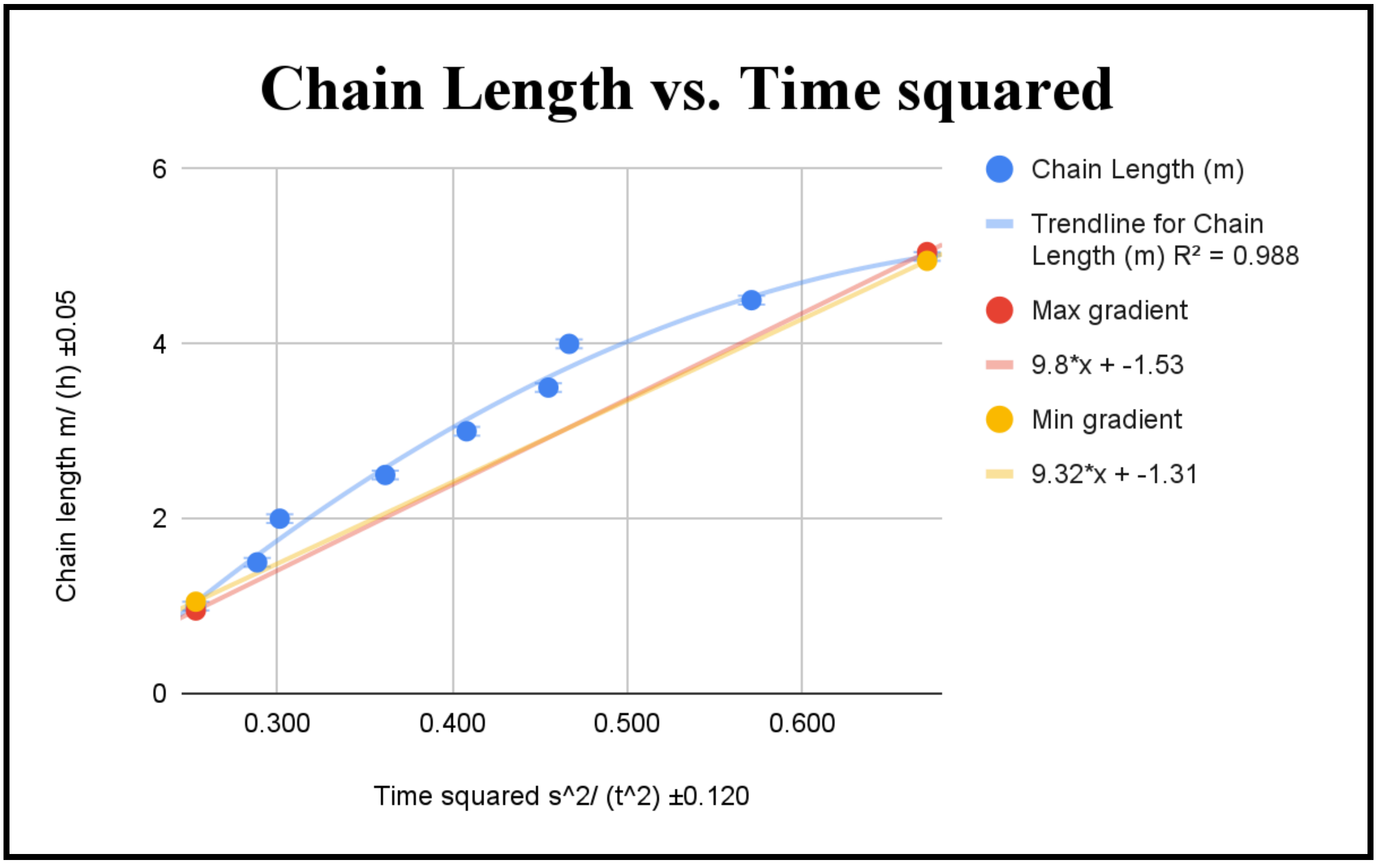
Height of Black Ball m/ (h) $\pm\ 0.05$	**Time taken to fall** $\pm\ 0.06$	**Actual time taken to fall** $\pm\ 0.06$	**Time squared** s^2/ **(t^2)** $\pm\ 0.12$	**Acceleration** ms^{-1}**/ (a)**
5.00	16.56	1.035	1.071	9.335
4.50	14.58	0.911	0.830	10.844
4.00	14.80	0.925	0.856	9.350
3.50	12.99	0.812	0.659	10.617
3.00	12.59	0.787	0.619	9.687
2.50	12.06	0.754	0.569	8.795

2.00	10.37	0.648	0.420	9.526
1.50	8.72	0.545	0.297	10.100
1.00	7.52	0.47	0.221	9.054

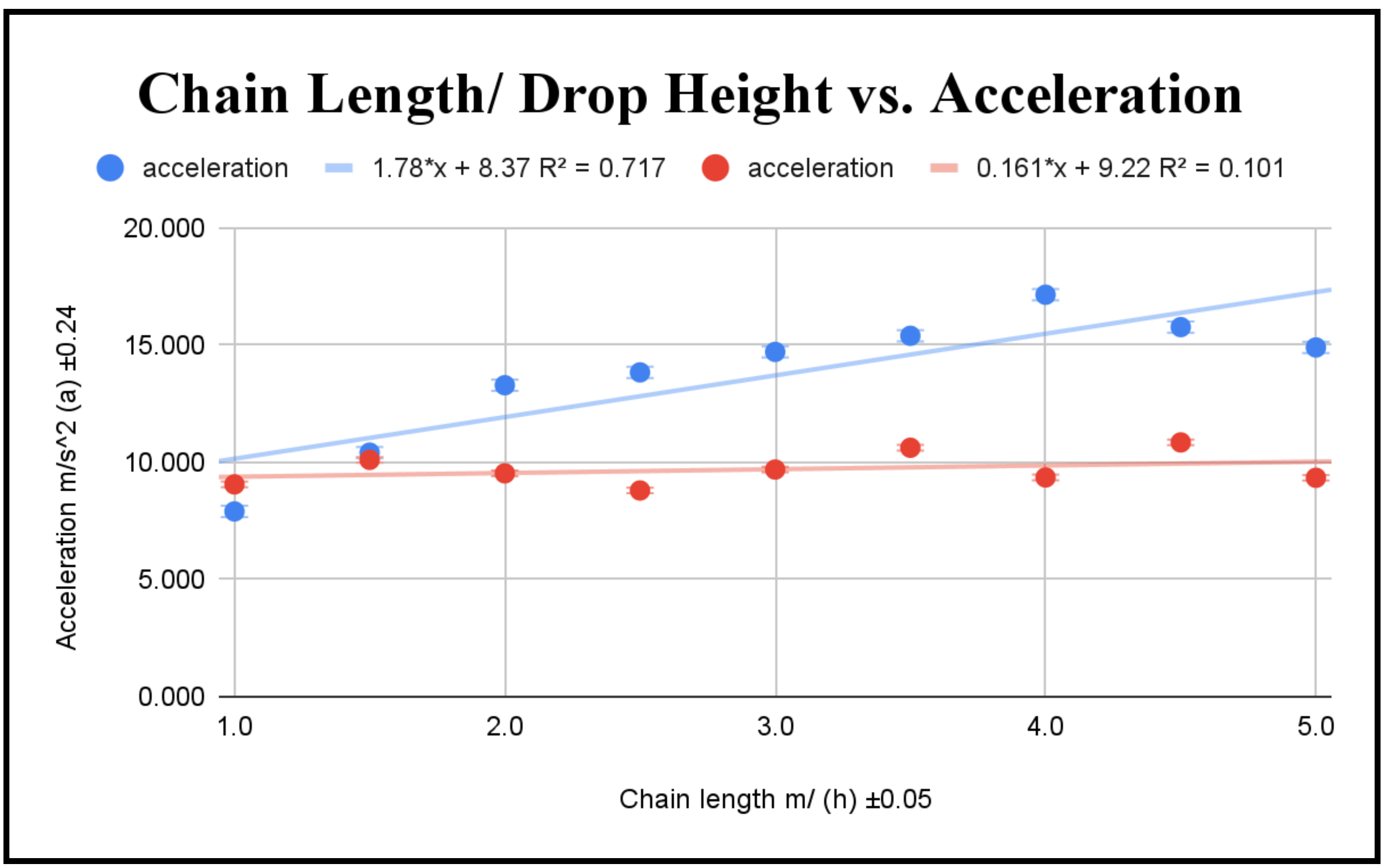
Graph 1.0 depicts the relationship between the height of the ball and the time taken for it to come in contact with the ground



Graph 2.0 depicts the relationship between time and length of the metallic chain



Graph 3.0 depicts the relationship between acceleration experienced by the sphere chain system and the freefall of the sphere.



Sample calculations

2) Average of time taken for the chain to unravel over 5 trials. $t_{average} = \frac{t_1+t_2+t_3}{3}$ $= \frac{12.73+13.54+13.08+12.92+13.29}{5}$ $= 13.11 \ \pm 0.06\, s$	Uncertainty propagation for average time taken ($t_{average}$) calculated using the special case of uncertainties when taking an average of given values. $\Delta m_{average} = \frac{\Delta t_1+\Delta t_2+\Delta t_3+\Delta t_4+\Delta t_5}{5}$ $= \ \pm \ \frac{0.06+0.06+0.06+0.06+0.06}{5}$ $= \ \pm 0.06$
2) Calculations to find t^2 value (Sample calculator for 5m chain) $t^2 = t \times t$ $= 0.819 \times 0.819$ $= 0.671 \pm 0.120\, s^2$	Uncertainty propagation for t^2 $\frac{\Delta t^2}{t} = 2\frac{\Delta t}{t}$ $= \frac{2\times0.06}{t} \times t$ $=\pm 0.120$
3) Gradient and acceleration (gravitational constant g) calculation for sphere is given by the 2 $\times$ the gradient (m) of the displacement vs. time squared graph through **Equation 3.0** in background information section $g = 2 \times gradient$ $g = 2 \times 4.88$ $g = 9.76 \ \pm 0.12\, ms^{-2}$	Uncertainty propagated for gradient and therefore acceleration is given by worst fit line method applied to **Graph 1.0** where $\Delta g = \frac{m_{max}-m_{min}}{2}$ $\Delta g = \frac{4.82-4.59}{2}$ $\Delta g =\pm 0.12$
4) Gradient and acceleration calculation for chain sphere	Uncertainty propagated for gradient and therefore

system for the data point (5, 0.671) given in Table 2.0 $s = \frac{1}{2}at^2$ $a = \frac{2s}{t^2}$ $a = \frac{2\times5}{0.671}$ $a = 14.895 \pm 0.240$	acceleration is given by worst fit line method applied to **Graph 2.0** where $\Delta g = \frac{m_{max} - m_{min}}{2}$ $\Delta g = \frac{9.8 - 9.32}{2}$ $\Delta g = \pm 0.24$

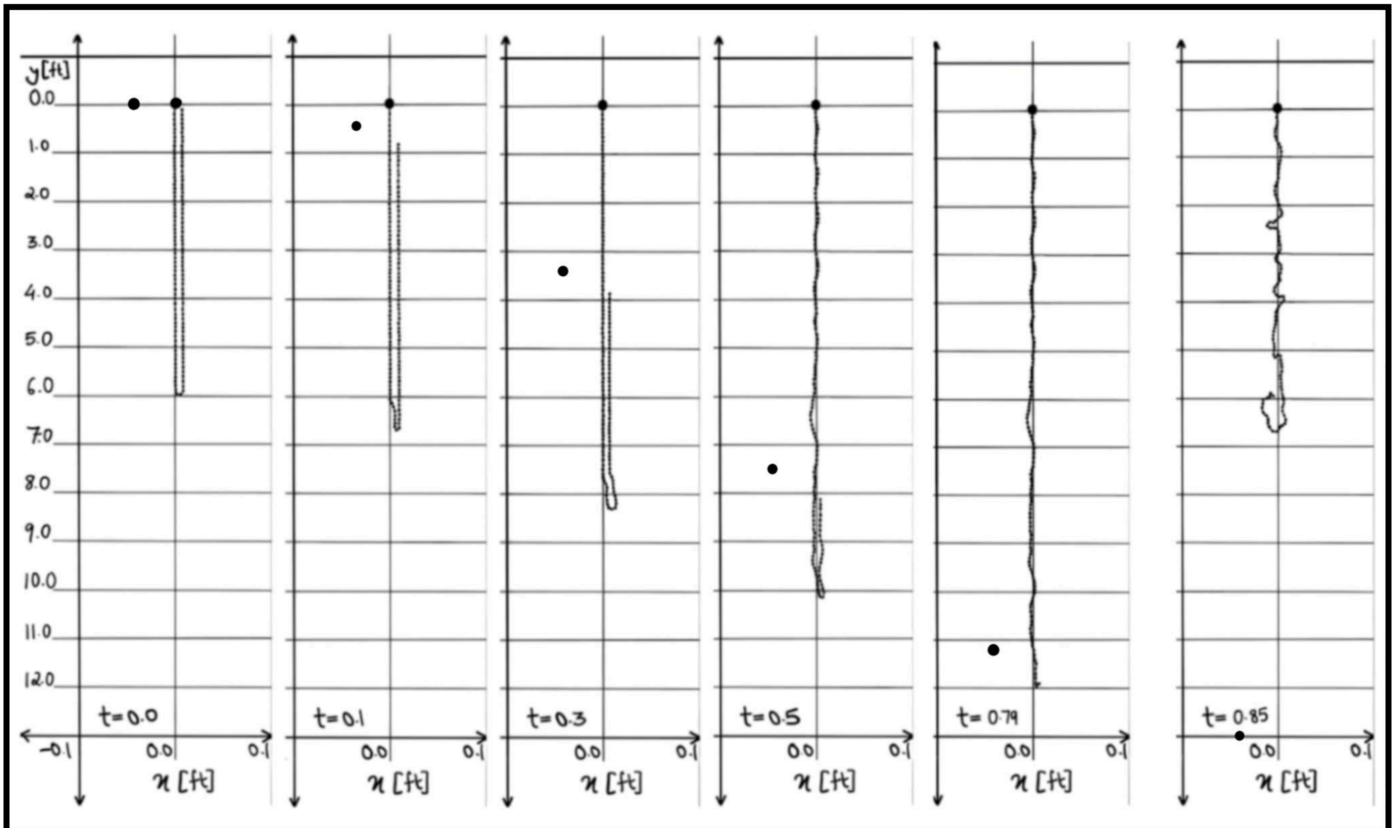
Error Percentage calculation

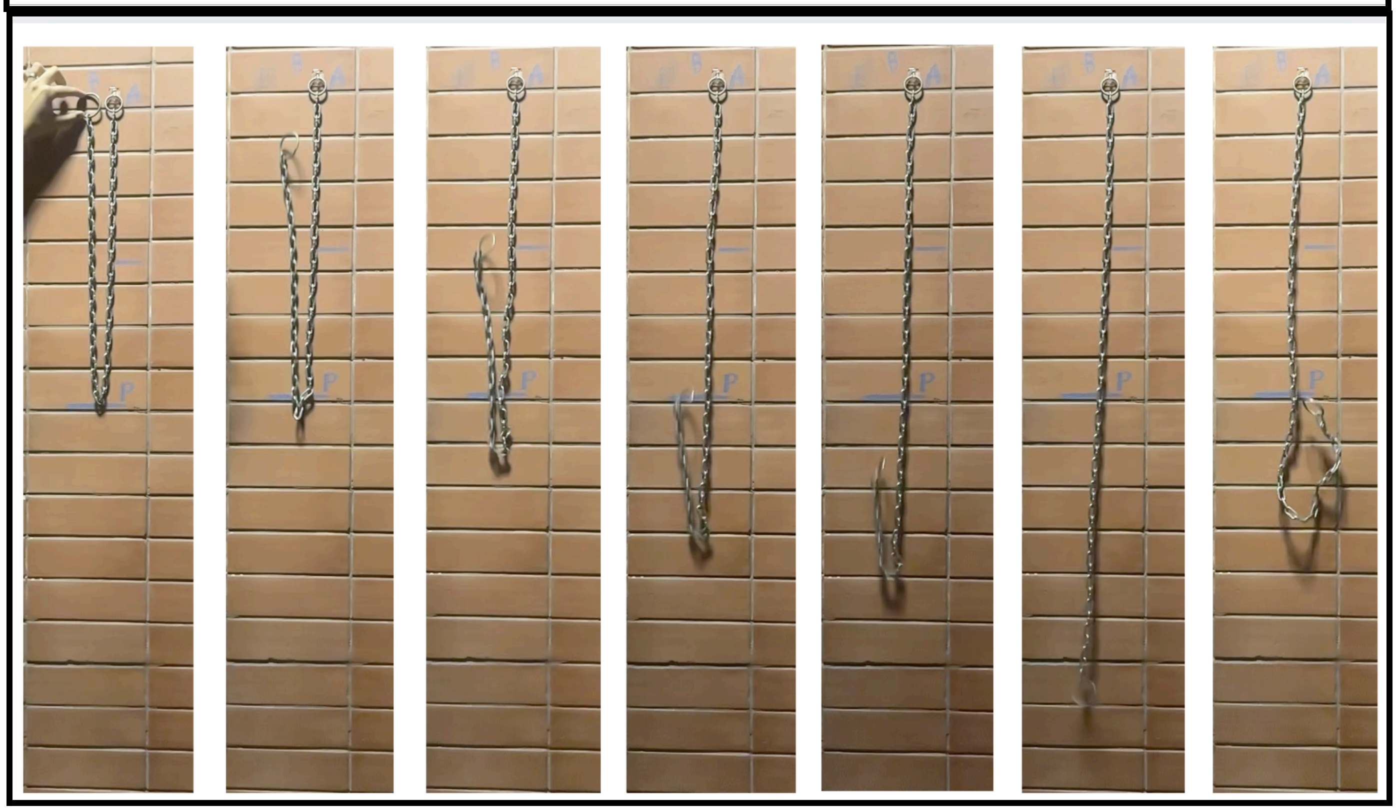
We can determine the error percentage for the experiment by comparing the experimental value of gravitational acceleration and its literature value which is $9.81\ ms^{-2}$

$$Error\,\% = \left|\frac{experimental\ value - literature\ value}{literature\ value}\right| \times 100$$

$$Error\,\% = \left|\frac{9.76 - 9.81}{9.81}\right| \times 100$$

$$Error\,\% = 0.51$$





Graph 4.0 visually depicts the dynamics of the unravelling of the chain of length 12 feet.

9.0 Conclusion

Graph 3.0 depicts how for the chain sphere system the value of acceleration was not constant but rather kept increasing with an increase in length of the chain given by the strong correlation coefficient of 0.717 highlighting a strong positive relationship between the two variables. The relatively low R^2 value for that graph can be justified due to the dynamic unraveling nature of the chain. Unlike free fall motion, the chain did not accelerate uniformly from the moment it was released. Instead, there was an initial lag as the chain unfolded during which the acceleration was lower. As more of the chain was released the acceleration increased deviating from the assumption we made that the acceleration was constant. This variation in acceleration throughout the motion led to inconsistencies in **Graph 3.0** reducing the linearity of the graph. However, through **Graph 3.0** it's evident that all the values of acceleration of the unravelling chain were recorded to be higher than the free fall of the same sphere. This implies that for the same distance travelled by the end of the chain and the sphere the chain was faster than the sphere along when no external force acting (gravity only). Moreover the graph shows that the vertical distance between the graphs increases with increasing length which implies that longer chains experience larger accelerations but counterintuitively take longer to unravel which can be justified due to larger mass of chain under the sphere of suspension. Therefore the characteristics found in **Graph 3.0** further strengthens the relationship deduced between length of the chain and the time taken for the chain to unravel. There are multiple outliers evidently seen in **Graphs 1.0 , 2.0 and 3.0** that have been outlined in **Section 10.0** below. However, despite said outliers we deduced the error % of the experiment to be as miniscule as 0.51%. Although only the value for the gravitational acceleration constant was compared to its literature value due to unavailability of literature values for time taken for a chain to unravel based on its length since the ball was dropped in free fall in similar conditions along with the chain sphere system we can assume it to be the error % for the experiment itself. The small error value ensures the reliability and replicability of the experimental procedure.

From the equation of the best fit line. As mentioned in the background information we assume acceleration due to gravity on the sphere to be $9.8\ ms^{-1}$ by travelling the same distance in a smaller time the end of the chain accelerates faster than the sphere. This implies that the chain did not only have a larger velocity but rather had a larger rate of change of velocity. The visual representation of the recorded motion as shown in **Graph 4.0** which is an outline of the video analysis of the subsequent motion of the ball and chain makes the hypothesis explicit and true as it's clearly evident that the chained ball reaches the displacement mark before the lone ball , and therefore has covers the same distance in a shorter interval of time than the lone sphere. The minimal error percentage between the calculated theoretical value obtained through theoretical evaluation of the situation in the background theory section strengthens the argument of the hypothesis and deems the methodology and procedure to be accurate as shown in the sample calculation **Table 5.0**.

Therefore it can be concluded that while a longer chain takes more time to unravel it does so with increasing acceleration.

10.0 Evaluation

10.1 Strengths

10.1.1: The lone ball was dropped with the chain in each trial to account for any weather disturbances reducing random errors and improving precision of the experiment

10.1.2: The clamp ensured that the chain and ball were dropped from the same height in every trial as using human hands to place and drop the chain and ball would've caused significant errors in reaction time and placement after adding links to the chain.

10.1.3: I initially used tape to mark Point A, B and P and the increments along with the lowest point when the chain has unravelled entirely but i realised that the thickness of the tape would hinder when i would stop the timer, and therefore i used chalk and soft pastels to mark the increments of distance and the necessary crucial points.

10.1.4: By slowing down the recording of the motion of the ball chain system human error was reduced and the time recorded through the video camera by pausing at the frame when the ball reached the finish mark was more precise than time that would've otherwise been recorded using a stopwatch.

10.2 Weakness

10.2.1: The lengths of the chain chosen were long and by evaluating it for unravelling its entire length instead of for a shorter distance the effect of air resistance is said to be larger and its effect on the chain's dynamics can be seen in **Graph 4.0** where as the chain unravels it becomes less stable and isn't linear. This air resistance was not accounted for in the time taken for the chain to unravel, which caused a systematic error as depicted by the y intercept of the Graph 1.0 and 2.0. (graph not passing from origin is a sign of systematic error). **Improvement:** Instead of recording the time taken by the chain to completely unravel from its folded state to a hanging state, the time taken for the chain to reach the lowest point when suspended in its U folded nature could be considered. Since the system would travel a shorter distance the random variation in the chain would reduce. This can be supported by Graph 4.0, evidently showing how the chain falls and comes to rest for 0.3s after which the chain becomes unstable as the tension begins to vary and isn't at rest anymore due to which there are tremors throughout the chain.

10.2.2: The outliers seen in Graph 1.0 and 2.0 were due to noticeable gusts of wind due to the outdoor setting of the experiment causing the generation of random errors. This could've been minimized by choosing a more controlled indoor environment but drilling nails indoors was prohibited due to which there was no feasible alternative to reduce the impact of unpredictable weather conditions. **Improvements:** Choosing a smaller displacement interval could've reduced the effect of gusts of wind. Furthermore the chain sphere system could've been encapsulated in a transparent draught shield that would've blocked out gusts of winds from affecting the chain's motion.

10.2.3: Since the chain was secured using a nail and was freely suspended from said nail there could've been shifting and rotation about the nail as the chain was not completely fixed. Moreover due to constant experimentation and long duration of hang time of the chain, the nail could've bent and lowered the point A impacting the results causing random errors. **Improvements:** The chain could've been attached to the wall using more than one nail. Moreover a rope could've replaced the chain which would have reduced the weight hanging from the nail decreasing the impact of this error.

10.2.4: For the experiment we assumed that the rope/chain was infinitely extensible however the chain was made up of links that rub together during the unravelling and might release mechanical energy in the form of heat. **Improvements:** replacing the chain with a rope would've decreased this error. Furthermore experimenting with the chain itself could've been improved by adding oil to the chain links decreasing the coefficient of friction and hence the loss of energy.

10.2.5: By using the equation of motion $s = \frac{1}{2}at^2$ the assumption that acceleration was constant throughout the motion of the chain sphere system was made, which is largely inaccurate as the number of chain links under the sphere account for what drags the ball vertically lower toward the ground. **Improvements:** The first two derivations done in the background information section of this experimental study take into account more factors such as the variability in acceleration of the chain sphere system. Moreover catenary effect in chains could've been considered to improve the analysis of the dynamic motion under study.

10.2.6: The sphere chain system used to find the gravitational acceleration constant had a noticeable volume and mass that couldn't be assumed to have experienced zero drag force that would've affected the accuracy of the experiment.

Improvements: Stokes law for the drag force acting on spherical bodies could've been taken into account while considering the net force acting on the system. Otherwise a smaller, lighter object could've been attached to the end of the chain or the last link of the chain could've been coloured to recognise during video analysis reducing the systematic error generated by ignoring air resistance.

11.0 Extensions

- One plausible extension to the above experiment that could produce more relevant results could be incorporating a stretch rubber like material and assessing the height reached by the mass attached at the end of the rubber string while considering catenary effect of the rope. This extension would require the consideration that the spring force in the rubber string would influence the unravelling length. This extension could be applied to real life scenarios such as bungee jumping, space elevator tethers, suspension bridges, spinal cord biomechanics, marine engineering for finishing and many more.

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